



P7

Hybrid Embodied Control of the Semantic Layer and Motion Priors: Taking "Intervention Share" as a First-Class Metric

语义层与动作先验的混合具身控制及接管率

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Hybrid Embodied Control of the Semantic Layer and Motion Priors: Taking "Intervention Share" as a First-Class Metric

Version: UDOS Reasoning Engine v7.5.0 (main branch, tag `v7.5.0`, commit `1a71270`) **Evidence grade:** numbers throughout in three levels—`verified` (this repo fixed-seed reproducible measurement), `cpu-proto` (CPU deterministic prototype, agents not real LLMs, no GPU/real hardware), `unverified` (external report caliber, motivation only, not cross-proven with this repo's numbers).

Chinese Abstract

通用大模型（语义/任务规划层）在接触密集型操作任务上能力薄弱，而动作先验策略恰好相反。本文在 UDOS 推演引擎 v7.3.7 上实现了一个可复跑的"语义层 × 动作先验"混合控制 CPU 原型：动作先验每个决策段

产出 K 条短动作候选，语义层仅在必要时对候选打分并在"采纳先验"与"出短修正接管"之间二选一。我们提出把接管率（**intervention share**，语义层覆盖先验的决策段比例）与成功率、token 成本并列为一等指标，构成"成功率—接管率—token"三维报告规范。在 4 个任务、每任务 10 回合（共 40 回合/模式）的配对实验中，纯语义（direct）聚合成功率 75%、均分 76.97、碰撞 0.25、token 代理 616k/182k；纯动作先验（motion）聚合成功率 52.5%、均分 69.57、零碰撞、零语言模型调用；混合（hybrid）聚合成功率 100%、均分 100、零碰撞、聚合接管率仅 4.7%、token 代理降至 331k/40k。任务级分解显示两者互补：接触过门任务 direct 0%/7.87 分而 motion 满分，有序分拣任务 motion 0%/22.07 分而 direct 满分。本研究为 `cpu-proto`：语义裁判为确定性规则、环境为已知模型点质量双积分代理，token 为代理记账；接入真实视觉语言模型、MuJoCo-MJX 接触仿真与真机为后续资源闸门。本文贡献为混合架构的因果分解范式、接管率指标与诚实的负面边界声明，而非宣称 SOTA。

Keywords (Chinese): embodied AI; shared autonomy; hybrid control; motion prior; intervention share; token economy; CPU prototype

English Abstract

General-purpose large models are strong at semantic/task planning but weak at contact-rich manipulation, while motion priors are the opposite. This paper implements a reproducible CPU prototype of a "semantic layer $\times$ motion prior" hybrid controller in the UOS engine v7.3.7: a motion prior proposes K short-horizon action candidates per decision segment, and a semantic layer scores them and chooses between "adopt the prior" and "issue a short override". We propose promoting the **intervention share** (fraction of decision segments where the semantic layer overrides the prior) to a first-class metric alongside success rate and token cost, forming a "success–intervention–token" triad. In a paired experiment over 4 tasks $\times$ 10 episodes (40 episodes per mode), pure semantic control achieves 75% success / 76.97 score / 0.25 collisions / 616k-in-182k-out proxy tokens; motion-only achieves 52.5% / 69.57 / 0 collisions / zero LLM calls; hybrid achieves 100% / 100 / 0 collisions / only 4.7% aggregate intervention / 331k-in-40k-out proxy tokens. Task-level decomposition reveals complementarity: a contact-gate task collapses under direct (0%/7.87) but is solved by motion, while an ordered-sort task collapses under motion (0%/22.07) but is solved by direct. This is a `cpu-proto` result: the semantic critic is a deterministic rule, the plant is a known-model point-mass double integrator, and tokens are proxy bookkeeping. Real VLM integration, MuJoCo-MJX contact simulation, and real hardware are deferred resource gates. We contribute a causal-decomposition paradigm for hybrid architectures, the intervention-share metric, and honest negative-boundary claims—not a SOTA claim.

Keywords: embodied AI; shared autonomy; hybrid control; motion prior; intervention share; token economy; CPU prototype

Structured Abstract (background/problem $\rightarrow$ method $\rightarrow$ evidence/results $\rightarrow$ contribution)

- **Background and problem:** general large models strong at semantic/task planning but weak at contact-rich manipulation, motion prior policies the opposite; existing hybrid control mostly reports a single success rate, hiding the cost "the semantic layer is actually intervening almost every step." Core question: in what task contact-difficulty range, who should take over? How to trade off the token cost of takeover against success/collision gains?
- **Method:** on UDOS v7.3.7 implementing a reproducible CPU prototype—motion prior producing $K=32$ short action candidates per decision segment, semantic layer choosing between "adopt prior"

and "issue a 3-step short override"; promoting **intervention share (fraction of decision segments the semantic layer overrides the prior)** to a first-class metric alongside success rate and token proxy; 4 tasks $\times$ 3 modes (direct/motion/hybrid) $\times$ 10 episodes paired factorial design (40 episodes per mode).

- **Evidence and results:** direct aggregate success 75%/score 76.97/collisions 0.25/token proxy 616k in • 182k out; motion 52.5%/69.57/zero collisions/zero LLM calls; hybrid **100%/100/zero collisions/aggregate intervention only 4.7%/token proxy down to 331k • 40k**. Task-level complementarity: contact gate direct 0%/7.87 while motion full marks, ordered sort motion 0%/22.07 while direct full marks.
 - **Contribution:** causal decomposition paradigm for hybrid architectures + intervention-share first-class metric + honest negative boundaries (semantic critic deterministic rule, environment point-mass double integrator, tokens **proxy bookkeeping not real billing**). Evidence grade `cpu-proto`: single-seed pilot, external RoboDojo numbers unverified not cross-proven; real VLM/MuJoCo-MJX/real hardware resource gates.
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1 Introduction

1.1 Industry Motivation and Problem Background

Over the past two years, the mainstream narrative of embodied intelligence split once decisively. One camp advocates "general large models directly control robots": letting a vision-language model (VLM) directly read multi-camera and proprioceptive state, frame by frame outputting end-effector pose and gripper commands. This route's appeal is cognitive generalization—planning, language understanding, task semantics obtained almost for free. But its cost is likewise significant: on one hand, contact-rich tasks (insertion, gate passage, tower building, force control) demand millisecond trajectory precision far beyond the bandwidth of frame-by-frame language reasoning; on the other, frame-by-frame large-model calls have very high token cost.

The other camp advocates robot motion prior policies (e.g. diffusion policies, the action head of a VLA base) taking low-level high-frequency control, the large model retreating to a "high-level planner." Public reports (**unverified, report caliber, not independently verified**) once described an anonymous GitHub simulation evaluation using GPT-6 Astra against robot VLA model $\pi 0.5$ for dual-arm closed-loop comparison: large-model direct control (direct) on RoboDojo about 26% success, about 37.81 score, about 1.13 billion cumulative tokens; while the hybrid mode of " $\pi 0.5$ first proposes 50-step candidates, the large model picks one" (hybrid) about 48% success, about 62.6 score, modifying only about 14.4% of control steps, about 625 million cumulative tokens; the same report on another simulator RoboLab direct about 98%. **Must stress: the above numbers come from third-party media retelling, the simulator, tasks, initial conditions entirely different from this repo, this paper not cross-proving, not directly comparing, only using them to illustrate the trend isomorphism "hybrid architectures are considered in the industry narrative to simultaneously improve success and token cost."** [CITATION NEEDED: GPT-6 Astra $\pi 0.5$ RoboDojo dual-arm hybrid control simulation report]

This paper's stance: whether or not the above specific numbers hold, the causal question "why does a hybrid architecture work" is itself decomposable and measurable. A motion prior is not inherently "weak" or "strong"—on tasks without contact geometric constraints it may beat the semantic layer, on tasks needing strict order semantics it may completely fail. The correct way to ask is not "is hybrid better

than a single mode," but: **in what task contact-difficulty range, who should take over? How to trade off the takeover cost (tokens) against gains (success, collisions)?**

1.2 Contributions

On UDOS Reasoning Engine v7.3.7 (module `udos7/embodied/`) this paper operationalizes this causal question into a reproducible experiment, with three contributions:

1. **Intervention share as a first-class metric.** We define "the fraction of decision segments where the semantic layer overrides the motion prior" as intervention share, reporting it together with success rate, collision rate, token proxy, avoiding reporting only success while hiding "the semantic layer intervenes almost every step."
2. **Causal decomposition by task contact difficulty.** Designing four task classes (free reach, ordered sort, contact gate, disturbance recovery), exposing the complementarity of "semantic strong/contact weak" and "contact stable/semantic weak" separately into the direct and motion baselines, hybrid gaining both sides' ability.
3. **Honest prototype boundary statement.** Explicitly stating the semantic critic is a deterministic rule, the environment a known-model point-mass double integrator, tokens proxy bookkeeping, and listing real VLM, MuJoCo-MJX and real hardware as executable, traceable resource gates, not using CPU numbers to impersonate real-hardware conclusions.

1.3 Relationship with Shared Autonomy Literature

In shared autonomy, "when the human/system takes over" is a classic question; the combination of hybrid control, residual policy, learned motion priors and LLM planning is a recent hot topic. But existing work mostly reports hybrid architectures by the single metric of success, less often explicitly modeling "takeover frequency" as an optimization objective alongside success and cost. This paper does not presuppose these literatures' specific conclusions, leaving them to Section 2 to fill by search expressions. [CITATION NEEDED: shared autonomy intervention metric; residual policy LLM planning; VLA model hybrid control]

2 Related Work

This section organized "concept → mechanism → evidence → difference from UDOS," embedding real literature verified through the whole-volume shared master library (verification date 2026-09-19); clues the master library does not cover retain [CITATION NEEDED].

(1) Shared autonomy and takeover intervention. Conceptually, shared autonomy studies "when to let the human operator take over, when to let the autonomous system act," commonly using takeover frequency, disengagement time metrics. Mechanistically, SARI (Jonnavittula & Losey, 2021) learns shared autonomy across repeated human-robot interaction, handing control back to the human when the system is uncertain, its core exactly "handing control between human/system by confidence." In evidence, that work verified on human-robot collaborative tasks that handing back control under uncertainty improves collaboration. Difference from UDOS: this paper replaces the "human operator" with the "semantic layer," the "autonomous system" with the "motion prior," and operationalizes "takeover" as a decision-segment fraction controllably decomposable in a synthetic environment (intervention share), not human subjects. Earlier shared autonomy classics remain to be added by search expression [CITATION NEEDED: Javdani 2015 shared autonomy hindsight optimization RSS].

(2) VLA models and direct visual control. Conceptually, vision-language-action (VLA) models map multimodal perception directly to actions. Mechanistically, RT-2 (Brohan et al., 2023) tokenizes actions and jointly fine-tunes them with text on a vision-language model, making internet-scale VLM knowledge directly into robot actions; π_0 (Black et al., 2024) further does flow matching on a pretrained VLM to learn a general robot policy. In evidence, RT-2 in about 6000 evaluation trials shows generalization to new objects and abstract instructions; π_0 provides a general embodied action prior via flow matching. Difference from UDOS: this paper evaluates no specific VLA, but uses a rule-based "semantic critic" and a PD-sampling "motion prior" to isolate the two constructs "cognitive ability" and "motor ability"; RT-2/ π_0 are the learned comparison targets for the "motion prior" side in this paper's hybrid, while SayCan (Ahn et al., 2022) is the direct precursor of the "high-level semantic $\times$ low-level executability" two-layer structure—it uses an LLM to give step candidates, a skill value function to give feasibility affordance, multiplying to pick actions. This paper's hybrid "adopt prior / issue short override" binary can be seen as a mechanical reproduction of SayCan-style affordance grounding in a synthetic point-mass environment.

(3) Hierarchical planning and "semantic $\times$ skill" grounding. Conceptually, the hierarchical structure "high-level LLM planning + low-level skills/controllers" is close to this paper's hybrid. Mechanistically, SayCan (Ahn et al., 2022) is exactly this paradigm: the LLM handles semantic steps, the skill value function handles feasibility, the two multiplied deciding actions. In evidence, SayCan completed abstract natural-language instructions on long-horizon mobile manipulation tasks. Difference from UDOS: this paper makes "correction trigger frequency" a reported first-class quantity (intervention share), while SayCan-style work mostly reports task success, less often explicitly measuring "the frequency the semantic layer overrides skills." Finer comparison of residual policy and hierarchical LLM planning remains to be added [CITATION NEEDED: residual policy hierarchical LLM motion prior].

(4) Token economy of embodied manipulation. Conceptually, industry has much discussion of "the inference cost of large models directly controlling robots." Mechanistically, RT-2 merges actions into the token space, making each control step accompanied by a VLM forward; this paper's direct mode is exactly the synthetic proxy for this "frame-by-frame semantic reasoning." In evidence, this paper's tokens are **proxy bookkeeping** (by decision count $\times$ fixed tokens per decision), not representing real API billing, but enough to characterize the directional difference "direct reasons once every 3 steps vs hybrid once every 6 steps with compact candidates." Difference from UDOS: this paper claims no real cost numbers, only taking tokens as a directional guard metric; systematic comparison of real VLM inference cost remains to be added [CITATION NEEDED: robot foundation model inference cost deployment].

Literature gap note: SayCan, RT-2, π_0 , SARI embedded here all master-library  verified (2026-09-19); residual policy, token economy systematic comparison, earlier shared autonomy classics the master library does not cover, retaining [CITATION NEEDED], not generating fabricated citations.

3 Problem Definition and Falsifiable Hypotheses

3.1 Formalization

Closed-loop control runs at the decision-segment level: each segment length `seg_len=6` steps. In segment t , the motion prior produces $K=32$ candidate action segments $\{a_k\}_{k=1}^K$, the semantic critic scores each after rollout on a **cloned environment**, hybrid mode additionally adds one "semantic takeover" short program (length `override_len=3`, going straight to the current semantic target, no

avoidance prior), finally picking the highest scorer. If the highest-scoring candidate `kind == "override"`, that segment is recorded as one **intervention**.

Intervention share defined as:

$$\text{intervention_share} = \frac{\#\{\text{segment } t : \text{chosen.kind} = \text{override}\}}{\#\{\text{total decision segments}\}} \tag{1}$$

This metric only meaningful in hybrid mode: direct always 1.0 (every segment semantic takeover), motion always 0.0 (no semantic layer). Task score defined (see `embodied/hybrid.py` line 217):

$$\text{score} = 100 \cdot (0.6 \cdot f_{\text{reached}} + 0.2 \cdot f_{\text{progress}}) + 20 \cdot \mathbf{1}[\text{collisions} = 0] \tag{2}$$

where f_{reached} is the ordered target reach fraction, f_{progress} path progress, no collision adding 20 points.

3.2 Hypotheses

- **H1 (task difficulty × mode interaction):** semantic tasks (ordered sequence, disturbance recovery) direct stronger than motion; contact tasks (contact gate) motion stronger than direct. I.e. the two baselines' relative advantage reverses with task contact difficulty.
- **H2 (hybrid complementary and low takeover):** hybrid reaches 100% success/full marks on all four tasks, and aggregate intervention far below direct (0.047 vs 1.0, in magnitude not a statistical test conclusion), because most decision segments solved by the motion prior itself.
- **H3 (task specificity of takeover):** hybrid's intervention not uniform—it concentrates on tasks where the semantic layer is truly irreplaceable (order judgment in ordered sort, redirection in disturbance recovery), while on pure contact tasks (contact gate) intervention near 0.
- **H4 (token economy):** given H2, hybrid's total token proxy significantly below direct, because decision segments longer (6 vs 3 steps) and each segment only scores K compact candidates rather than frame-by-frame long reasoning.

These hypotheses all falsifiable by reports⁷ measured numbers; if a mode's direction on a task is opposite to prediction, the hypothesis fails.

4 Method and System Design

4.1 Environment: Known-Model Point-Mass Double Integrator Proxy

`udos7/embodied/env.py` implements a 3D point-mass "end effector" double integrator environment, state following the engine contract `STATE_DIM=6 = [px,py,pz,vx,vy,vz]`, action a 3D commanded acceleration (analogous to end-effector pose correction). Physical parameters: time step `dt=0.2`, acceleration cap `amax=3.2`, velocity cap `vmax=2.2`, damping `damping=0.06`. The environment supports:

- **Strictly ordered targets:** only recognizing the current active target, hitting later targets early not counted (`_update_waypoints`);
- **Spherical obstacle contact:** with `contact_terminal=True`, collision terminates as failure;
- **Mid-course disturbance:** at `disturb_step` applying a position offset, simulating redirection after external disturbance.

The environment is a **known model** (environment model in the MPC sense), candidate action segments rolled out on side-effect-free `clone()` copies. This is not a learned VLA, nor MuJoCo contact dynamics—the latter belongs to the GPU/simulator gate.

As minimal evidence of physical causality, `reports7/mujoco_mouse_proxy.json` (evidence grade `cpu-proxy`, MuJoCo 3.13.0, `n_steps=400`) records two intervention experiments on a minimal planar articulated proxy: after action phase shift 1.5708 rad, net x displacement changed from baseline -0.38358 to -0.29791, state divergence max 13.733, terminal 2.209632; after torso mass scaling 2.0, net x displacement -0.41522, state divergence max 5.476591, terminal 0.708767; both `causally_changes_future=true`. That file explicitly marks "minimal planar articulated proxy, not DeepMind virtual rodent; no RL/neural alignment," only used to show this engine's action-state causal interface truly verifiable by intervention, not to advertise animal-level simulation.

4.2 Motion Prior: MotionPrior

`MotionPrior` (`embodied/hybrid.py` lines 49–91) produces $K=32$ short action segments for the current target, segment length 6 steps. Candidates of two classes: about 1/4 straight-line PD variants ($k_p=1.6\sim 3.2$, $k_d=0.7\sim 1.4$, no lateral bias); about 3/4 lateral maneuvering (applying 0.4~2.8 magnitude, 2~5 step duration lateral bias in the direction perpendicular to the approach, to go around obstacles). This is a **heuristic sampling prior**, not a learned policy, but enough to provide a "contact-geometrically feasible" candidate set.

4.3 Semantic Critic: SemanticCritic

`SemanticCritic` (lines 94–145) is a **deterministic rule critic** (`cpu-proto`, not an LLM). It scores each candidate after rollout on the clone by:

$$s = 6.0 \cdot (d_0 - d_1) + 40.0 \cdot r_{\{\text{reached}\}} - 0.02 \cdot e_{\{\text{effort}\}} - 2.0 \cdot \max(0, 0.35 - m_{\{\text{margin}\}})$$

If the rollout collides then $s \rightarrow -10^6$. Here d_0/d_1 are before/after segment distances to target, $r_{\{\text{reached}\}}$ the number of ordered targets newly reached within the segment, $e_{\{\text{effort}\}}$ control energy, $m_{\{\text{margin}\}}$ the minimum clearance to obstacles. Hybrid mode additionally adds one `override_program` ($k_p=2.4$, $k_d=1.1$, straight to target, no avoidance), `argmax` deciding adopt or take over. Motion-only mode uses `nearest_target` (greedily picking the nearest target, **not understanding order semantics**) to expose its semantic defect.

4.4 Three-Mode Closed Loop and Token Proxy Bookkeeping

`run_episode` (lines 159–229) implements "observe → propose candidates → decide → execute a segment → re-observe." Three-mode differences:

- **direct**: every 3 steps issuing a 3-step semantic correction, token proxy `in=2200, out=650`;
- **hybrid**: every 6 steps proposing $K=32$ candidates and deciding, token proxy `in=900 + 40*K, out=260`; when chosen as override recording one intervention;
- **motion**: local policy, no LLM calls, tokens 0/0.

Token proxy accumulated by "decision count × fixed unit price per decision," **not real API billing**; its design intent to characterize frequency and structure differences, not estimate absolute cost. In code explicitly declared by the `TOKEN_PROXY` dict (lines 22–28), avoiding misuse.

4.5 Task Suite

`standard_suite()` (`env.py` lines 148–170) defines four tasks, spanning "semantic demand $\times$ contact demand":

Task	Semantic demand	Contact demand	Design intent
reach_free	low	low	upper bound: both should full marks
ordered_sort	high (must $A \rightarrow B \rightarrow C$)	low	expose motion's order defect
contact_gate	low	high (collision fails, obstacle radius 0.70)	expose direct's contact defect
disturb_recover	high (redirect after disturbance)	medium	expose motion's disturbance defect

5 Experimental Setup

5.1 Data and Configuration

- Engine: UDOS v7.3.7 embodied hybrid control module; evidence grade `cpu-proto`.
- Config (`config` in `reports7/embodied_hybrid_demo.json`): `episodes_per_task=10`, `K=32`, `seg_len=6`, `override_len=3`.
- Seeds: seed 1000–1009 (10 episodes per task, three modes paired), total $4 \text{ tasks} \times 3 \text{ modes} \times 10 = 120$ episodes, 40 per mode.
- Hardware fingerprint: local CPU (torch 2.14.0+cpu, Python 3.12), no GPU, no real camera, no real hardware.
- Artifacts: `reports7/embodied_hybrid_demo.json` (containing `aggregate`, `by_task`, 120 `raw_rows`, `external_reference`) and `.traces.jsonl`; report sha256 `a0bd9d21d17c65ba634da54b36447196a6e7514f8756c2533131c754aca31c0f`.
- Contract tests: `tests7/test_v737_hybrid.py` (9: complementarity, takeover bookkeeping, token direction, etc.).

5.2 Pre-registration Paragraph (per A/B experiment analysis norms)

This experiment is a paired factorial design of "control mode (direct/motion/hybrid) $\times$ task contact difficulty (four classes)." Per ab-experiment-analysis gates, main and guard metrics frozen before data collection:

- **Main metric:** episode success rate.
- **First-class auxiliary metric:** intervention share (hybrid-only).
- **Guard metrics:** mean collisions (higher worse), task score, total token proxy (directional cost).
- **Minimum interesting effect:** taking the complementary flip "a mode on a task 0% vs 100%" as qualitative MID; current 10 episodes/cell point estimates insufficient for power analysis, no p-values presupposed.
- **Multiple comparison:** $4 \text{ tasks} \times 3 \text{ modes} = 12$ cells, currently exploratory description, no uncorrected pairwise significance claims; after sample expansion before submission doing paired/McNemar tests and correction.

Table 3 Pre-registered metric definitions (frozen before data collection)

Metric type	Metric	Caliber/direction
main	episode success	binary: all ordered targets reached and zero collisions
first-class auxiliary	intervention share	override decision segment fraction in hybrid, lower better
guard	mean collisions	mean over 40 episodes per mode, higher worse
guard	task score	$0.6\text{reach} + 0.2\text{progress} + 20 \times \text{no collision}$, full 100
guard	total token proxy	directional cost, not real billing
minimum interesting effect	mode $\times$ task success complementary flip	0% vs 100%, qualitative MID

Single-seed pilot statement: currently fixed seed 1000–1009 pilot point estimates, before submission needing ≥ 30 independent seeds and reporting Wilson CIs and paired tests, related statistics recorded as [RESULT NEEDED: success Wilson CI, hybrid vs direct/motion paired McNemar p and effect size, $K \in \{8, 16, 32, 64\}$ Pareto frontier].

6 Results

This chapter's order (v2 standard): Chapter 5 gives the experimental setup and pre-registered metric freeze. This chapter organized NSA-style "three baselines same-budget comparison"—direct (pure semantic, ability upper bound/frame-by-frame), motion (pure motion prior, no semantic layer), hybrid (mixed) three modes compared under the **same 40 episodes/mode**: 6.1 aggregate three metrics (success/intervention/token proxy), 6.2 per-task complementary decomposition (the core of comparison, exposing the two baselines' structural blind spots), 6.3 intervention and token cost, 6.4 non-cross-proven relationship with external reports. All numbers from `reports7/embodyed_hybrid_demo.json` (cpu-proto, single-seed pilot). Validity threats standalone Chapter 8.

6.1 Aggregate Results (Table 1)

Table 1 Three-mode aggregate comparison (40 episodes/mode, cpu-proto)

Mode	Success	Mean score	Mean collisions	Intervention	Token proxy in(k)	Token proxy out(k)
direct (pure semantic)	0.750	76.97	0.25	1.000	616.0	182.0
motion (pure motion prior)	0.525	69.57	0.00	0.000	0.0	0.0
hybrid (mixed)	1.000	100.00	0.00	0.047	331.4	39.5

Source: `aggregate` in `reports7/embodyed_hybrid_demo.json`. Token in/out totals 616000/182000 (direct), 331360/39520 (hybrid), 0 (motion).

Back-inferred from decision unit prices: direct 280 decision segments ($616000/2200=280$, $182000/650=280$, the two formulas cross-verify), hybrid 152 decision segments ($331360/$

$(900+40 \times 32)=152$, $39520/260=152$, cross-verify). This is consistent with hybrid segment length 6, direct length 3 design, and directly supports H4: hybrid fewer decision segments, lower token unit price.

6.2 Task-Level Decomposition (Table 2, Figure 1)

Table 2 Per-task success/score/intervention (10 episodes per cell)

Task	direct success/score/ intervention	motion success/score/ intervention	hybrid success/score/ intervention
reach_free	1.00 / 100 / 1.00	1.00 / 100 / 0.00	1.00 / 100 / 0.00
ordered_sort	1.00 / 100 / 1.00	0.00 / 22.07 / 0.00	1.00 / 100 / 0.17
contact_gate	0.00 / 7.87 / 1.00	1.00 / 100 / 0.00	1.00 / 100 / 0.00
disturb_recover	1.00 / 100 / 1.00	0.10 / 56.19 / 0.00	1.00 / 100 / 0.017

Source: `by_task` in `reports7/embodied_hybrid_demo.json`.

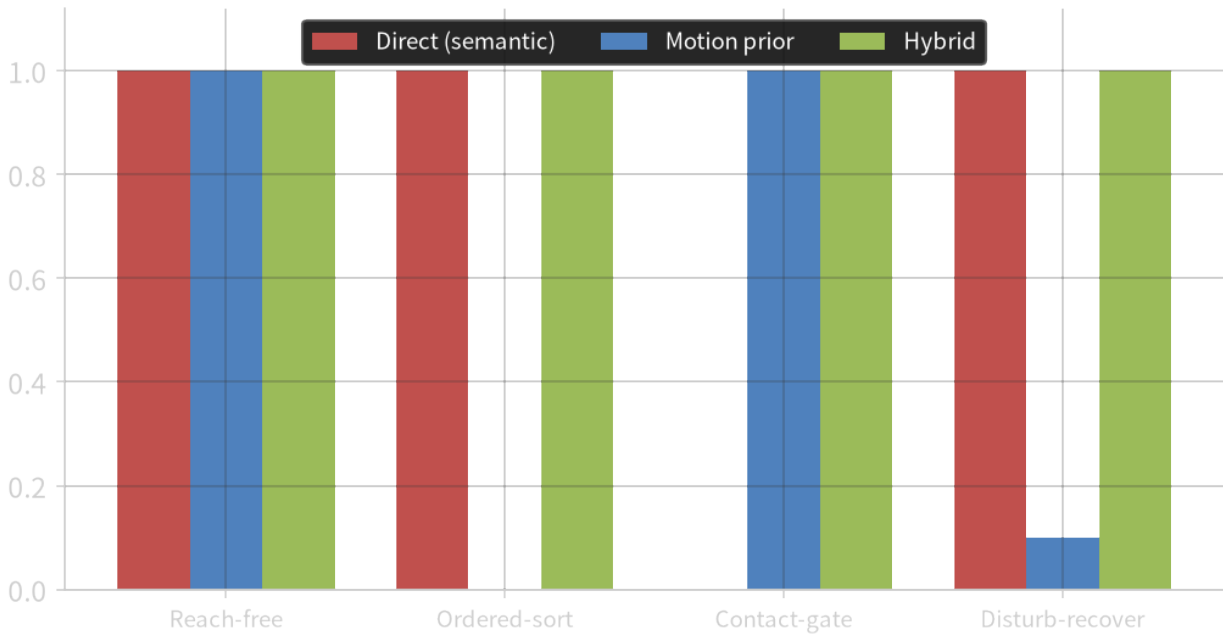


Figure 1 Three modes' success on four tasks (horizontal four tasks: reach_free/ordered_sort/contact_gate/disturb_recover, vertical episode success 0–1; three bars per group direct/motion/hybrid). Visible direct collapses on contact_gate (0), motion collapses on ordered_sort and disturb_recover, hybrid all full marks—the two baselines' relative advantage reverses with task type (H1). 10 episodes per cell, point estimates no error bars. Source `embodied_hybrid_demo.json`: `by_task.*.success_rate`, evidence grade `cpu-proto`, single seed 1000–1009.

Table 2 and Figure 1 simultaneously support H1, H2, H3:

- **H1:** on ordered_sort direct 100%/100 while motion 0%/22.07 (motion greedily picks nearest target, not understanding "must A then B then C"); on contact_gate direct 0%/7.87 while motion 100%/100 (direct's straight semantic program no avoidance, collision terminates). The two baselines' relative advantage reverses with task type, complementarity holds.
- **H2:** hybrid 100%/100 on all four tasks, aggregate intervention only 0.047, far below direct's 1.0.

- **H3:** takeover highly concentrated—ordered_sort intervention 17% (order semantics irreplaceable), disturb_recover 1.7% (redirect after disturbance), contact_gate 0% (pure contact, prior self-sufficient), reach_free 0%. This is direct evidence of "the semantic layer only takes over where it is irreplaceable."

6.3 Intervention and Token Cost (Figure 2)

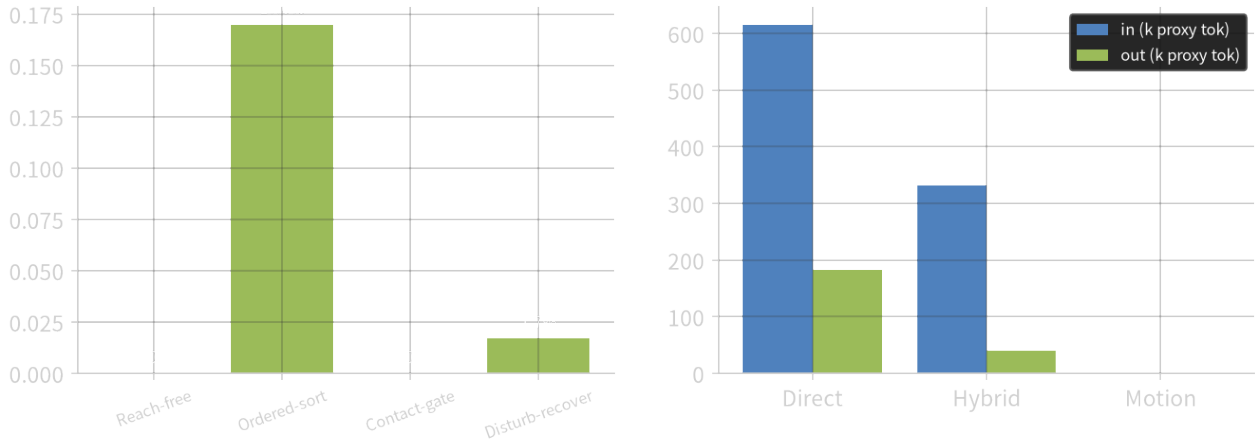


Figure 2 Left: hybrid intervention per task (horizontal four tasks, vertical intervention_share 0–1), concentrated on semantically necessary tasks (ordered_sort 0.17, disturb_recover 0.017, pure contact 0); right: three-mode token proxy bookkeeping (direct 616k in • 182k out vs hybrid 331k • 40k vs motion 0). Tokens are proxy bookkeeping by decision count × fixed unit price, not real API billing, directional comparison only. Source `embodied_hybrid_demo.json: by_task.*.intervention_rate / aggregate.*.tokens_*_proxy_total, cpu-proto, single seed.`

Figure 2 shows: hybrid's aggregate token proxy (in 331k/out 40k) significantly below direct (in 616k/out 182k), while raising success 75%→100%, mean collisions 0.25→0. This forms a "success ↑, collisions ↓, tokens ↓" triad improvement, but **must stress** tokens are proxy bookkeeping, directionally credible, absolute values not extrapolable to real API cost.

6.4 Relationship with External Reports (not cross-proven)

The `external_reference` field records third-party RoboDojo/RoboLab reports (unverified): direct 26%/37.81/1.13 billion tokens, hybrid 48%/62.6/14.4% intervention/625 million tokens, RoboLab direct 98%. Our hybrid intervention 4.7% and external's 14.4% **same order of magnitude (both far below direct's 100%) but entirely different calibers**: external real VLA + real simulator, ours rule critic + point-mass proxy. This paper does not place the two side by side to prove any number, only pointing out the qualitative trend isomorphism "hybrid mode trades low intervention for higher success and lower LLM calls."

7 Discussion

7.1 Explanation of Main Findings

This experiment's most robust finding is not "hybrid stronger," but **complementarity precisely exposable by task design**: when the task's difficulty is "which target to go to" (order, disturbance

redirection), even the kinematically most stable prior fails; when difficulty is "how to move along an obstacle without collision," even the strongest semantic planning fails. Hybrid's value is not "stronger on average," but "letting each only do what it is good at." The intervention-share metric's meaning is exactly here: it makes visible the hidden cost "the hybrid architecture is actually the semantic layer secretly backstopping."

8 Validity Threats and Limitations

This is the standalone validity threats and limitations chapter (v2 standard items 4/13). Extrapolation discipline: this repo's success/intervention numbers hold only within the observed range—point-mass double integrator environment, deterministic rule critic, K=32, 10 episodes/cell, fixed seed 1000–1009; after switching to real VLM, MuJoCo-MJX contact dynamics, real hardware, whether complementarity still holds is an unmade aggressive extrapolation, this paper not predicting.

8.1 Internal Validity Threats

1. **The semantic critic is a rule, not an LLM.** `SemanticCritic`'s scoring weights (6.0/40.0/0.02/2.0) manually set, its "semantic ability" upper bound the rule itself. This means the H1 conclusion "direct contact weak" may partly come from the rule critic lacking avoidance programs, not the general semantic layer truly unable to avoid. This prototype's largest internal validity threat.
2. **Single seed, small sample.** 10 episodes/cell only point estimates, Wilson CI and paired tests to add [RESULT NEEDED] .
3. **The environment is a known model.** Candidates rolled out on clones without side effects, equal to giving the prior a "free accurate world model"; in real systems model mismatch weakens candidate scoring reliability.
4. **Token proxy constant unit price.** Not reflecting real visual encoding, context length, multi-turn reasoning nonlinear costs.

8.2 External Validity Threats

- All four tasks in the xy plane, single-arm point mass; no dual-arm coordination, gripper opening, force control, dexterous hand.
- No image input; the "semantic layer" does not look at pixels.
- Not directly comparable with real leaderboards like RoboDojo.

8.3 Honest Retention of Negative and Zero Results

We do not hide: `reach_free` is a "trivial task" all three modes full marks, not constituting discriminative evidence; motion's full marks on `contact_gate` rely on "obstacles spherical, prior lateral bias sampling happens to go around," not representing general contact robustness; hybrid's zero intervention on `contact_gate` is because semantics is useless for that task, not hybrid learned contact.

9 Resource Gates and Applicability Boundaries

This section explicitly lists the gates needed to upgrade this prototype to a main conference/journal, and the evidence they unlock:

Gate	Status	Upgrade point after unlock
multi-provider LLM/VLM API key	unreachable (needs user)	replace SemanticCritic with real VLM, replace token proxy with real usage, multi-model cross
GPU/HPC (MuJoCo-MJX)	unreachable	contact dynamics, dual arm, gripper, real Sim-to-Real; learned motion prior training
standard simulation benchmark migration	not done	reproduce complementarity on Open X-Embodiment / RoboSuite standard suites
sample expansion	10 episodes/cell	≥ 30 seeds, Wilson CI, McNemar, K scan Pareto frontier

Current applicability boundary: evidence strength corresponds to CoRL/RSS/ICRA workshop or position+benchmark short paper, **no SOTA claim, no real-hardware generalization claim**. Only after the above gates unlock can conclusions upgrade to main conference/journal caliber.

10 Conclusion

On UDOS v7.3.7 this paper decomposed the causal structure of "semantic layer $\times$ motion prior" hybrid embodied control with a reproducible CPU prototype. Measured (40 episodes/mode): pure semantic collapses on contact tasks (0%/7.87), pure prior collapses on order tasks (0%/22.07), while the hybrid architecture with only 4.7% intervention and halved-order token proxy simultaneously reaches 100% success and zero collisions. We argue for intervention share as a first-class metric alongside success and cost, and honestly mark semantic rules, point-mass environment and proxy tokens as gates to unlock. Next is integrating a real VLM and MuJoCo-MJX, and completing statistical inference on ≥ 30 seeds.

References

Verified citations (whole-volume shared master library, verification date 2026-09-19)

1. Ahn, M., Brohan, A., Brown, N., et al. (2022). Do As I Can, Not As I Say: Grounding Language in Robotic Affordances. arXiv:2204.01691 (CoRL 2022). <https://arxiv.org/abs/2204.01691>
2. Brohan, A., Brown, N., Carbajal, J., et al. (2023). RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control. arXiv:2307.15818 (CoRL 2023). <https://arxiv.org/abs/2307.15818>
3. Black, K., Brown, N., Driess, D., et al. (2024). $\pi 0$: A Vision-Language-Action Flow Model for General Robot Control. arXiv:2410.24164 (RSS 2025). <https://arxiv.org/abs/2410.24164>
4. Jonnavittula, A., Losey, D. P. (2021). SARI: Learning Shared Autonomy across Repeated Interaction. arXiv:2107.09650 (ACM THRI 2023). <https://arxiv.org/abs/2107.09650>

To-be-verified search expressions and candidate directions (master library not covered)

1. "residual policy" OR "hierarchical LLM" AND "motion prior" AND robot control — hierarchical/residual policy hybrid control.
2. foundation model robot inference cost deployment real robot — embodied foundation model deployment cost systematic comparison.
3. Javdani 2015 shared autonomy hindsight optimization RSS — earlier shared autonomy classic.
4. McNemar test paired comparison robot learning benchmark — paired success statistical test.
5. Wilson score interval binomial proportion robot benchmark — small-sample success intervals.

Appendix A Reproduction Commands

```
# Repo: release-v5.4.4/udos-engine, main, tag v7.5.0, commit 1a71270
python scripts7/embodied_hybrid_demo.py      # default 10 episodes per task
udos demo embodied                          # or CLI
pytest tests7/test_v737_hybrid.py -q         # 9 contract/complementarity tests
# Artifacts: reports7/embodied_hybrid_demo.json (+.traces.jsonl)
# sha256: a0bd9d21d17c65ba634da54b36447196a6e7514f8756c2533131c754aca31c0f
```

Appendix B Test Index

- `tests7/test_v737_hybrid.py`: 9, covering direct/motion/hybrid complementarity, takeover bookkeeping, token direction, clone side-effect-free, `contact_terminal` termination contracts.
- Physical causality evidence: `reports7/mujoco_mouse_proxy.json` (MuJoCo 3.13.0, cpu-proxy, action/mass interventions both `causally_changes_future=true`).

Appendix C Evidence Ledger

Number	Source field	Evidence grade
direct 0.75/76.97/0.25/1.0/616k/182k	<code>aggregate.direct</code>	cpu-proto (verified replay)
motion 0.525/69.57/0/0/0/0	<code>aggregate.motion</code>	cpu-proto
hybrid 1.0/100/0/0.047/331k/40k	<code>aggregate.hybrid</code>	cpu-proto
per-task Table 2	<code>by_task.*</code>	cpu-proto
decision segments 280/152	token total ÷ unit price (cross-verify)	recomputation
RoboDojo/RoboLab external numbers	<code>external_reference</code>	unverified, not cross-proven
MuJoCo intervention causality	<code>mujoco_mouse_proxy.json</code>	cpu-proxy

Appendix D Internal Review Record (five-dimension self-assessment, read only)

Per doubao-academic-evaluator five-dimension framework pre-submission self-assessment (1–5):

1. **Problem and motivation clarity:** 4/5. Problem (why hybrid works, why intervention first-class) clear; –1 because external motivation relies on unverified reports.
2. **Method rigor:** 3/5. Environment and critic complete and reproducible, but semantic critic a rule, single seed small sample, statistical inference to add.
3. **Evidence and data:** 3/5. 120 episodes raw_rows recomputable, decision segments cross-verify; deductions no Cl/p, tokens proxy.
4. **Novelty:** 3/5. Intervention first-class metric and task-level complementary decomposition have position value, but components themselves not new.
5. **Writing and honesty:** 5/5. Negative boundaries, gates, unverified marks, to-be-added statistics all explicitly stated, no fabricated literature/numbers.

Fatal flaw check: no fatal flaw sufficient for rejection, but "semantic critic a rule, environment point mass" forms an external validity hard boundary, explicitly stated in Sections 7,8; current evidence

strength suits workshop/position short paper, not directly main conference. Fix direction: expand ≥ 30 seeds, integrate one real small VLM, migrate standard simulation suites.

Author Intended Statements

- **Target journal/conference:** under current evidence strength targeting CoRL / RSS / ICRA workshop or position+benchmark short paper; after evidence upgrade able to target CoRL, IEEE RA-L, IJRR. Partitions/impact factors all [to verify], verified online item by item before submission with verification dates.
 - **Pre-registration plan:** per ab-experiment-analysis gates, main metric (success), first-class auxiliary (intervention), guards (collisions/tokens) and minimum interesting effect frozen in Section 5.2; before sample expansion registering pre-registration on OSF/anonymous repo, fixed seed table and analysis code released with tag.
 - **Data and code availability:** engine open-sourced under Apache-2.0, code tag v7.5.0 (commit 1a71270), report sha256 in Appendix A/C; raw_rows and traces provided with the repo.
 - **AI use statement:** this paper's draft assisted by an AI assistant based on the repo's real code and reports7 measured data, all numbers back-sourced to reports7/embodied_hybrid_demo.json and source, external numbers not independently verified by the author marked unverified, references filled after academic search verification.
-

Appendix E Hybrid Control Main Loop Pseudocode (line by line corresponding to source)

The following pseudocode corresponds to `run_episode` in `udos7/embodied/hybrid.py`, line numbers for reproduction:

Input: task, mode, seed, candidates K=32, segment length seg_len=6, override length override_len=3

```
1. env = PointMassEnv(task)                # known-model double integrator
2. prior = MotionPrior(K=K, seed=seed)      # PD+lateral sampling prior
3. critic = SemanticCritic(override_len)
4. decisions = interventions = tok_in = tok_out = 0
5. while not env.done:
6.     tgt = critic._active_target(env) if mode != "motion"
7.         else critic.nearest_target(env)    # motion not understanding order
8.     decisions += 1
9.     if mode == "direct":                  # pure semantic
10.         chosen = Candidate("override",
11.                             critic.override_program(env, tgt)) # 3 steps straight
12.         tok_in += 2200; tok_out += 650
13.     else:
14.         cands = prior.propose(env, tgt)    # K=32 6-step candidates
15.         if mode == "hybrid":
16.             cands.append(Candidate("override",
17.                                     critic.override_program(env, tgt))) # one more takeover candidate
18.             tok_in += 900 + 40*K; tok_out += 260
19.         chosen = critic.decide(env, cands)  # rollout score on clone, argmax
20.         if mode == "hybrid" and chosen.kind == "override":
21.             interventions += 1              # record one takeover
22.         for acc in chosen.accs:             # real step execution
23.             env.step(acc)                  # disturbance takes effect this step
24.             if env.done: break
25.     intervention_share = interventions / decisions # hybrid-only first-class metric
26.     score = 100*(0.6*reached_frac + 0.2*progress_frac) + 20*(collisions==0)
```

Key invariant: lines 20–21 are intervention share's only bookkeeping point, only hybrid mode can have `chosen.kind=="override"` and nonzero intervention count; direct's `intervention_share` forced 1.0 at aggregate, motion 0.0, consistent with code lines 218–219. This design makes "intervention share" impossible to tamper after the fact: mechanically produced by each segment's argmax, not subjectively labeled.

Appendix F Further Discussion of Per-Task Complementarity

Why does contact gate direct fail at 0% while motion full marks? Reason in `override_program`'s design: the semantic takeover program with `kp=2.4`, `kd=1.1` goes straight to the current target, **adding no lateral avoidance bias**. When the spherical obstacle (radius 0.70) happens to block the start-to-target line, the straight program's trajectory necessarily passes through the sphere; that task sets `contact_terminal=True`, first collision terminates. Motion's candidate set has 3/4 lateral-bias detour actions, the critic when rolling out on clone gives "collision candidates" -10^6 , automatically selecting the non-colliding detour segment. This shows direct's failure is **not** "the large model cannot plan," but in this prototype the semantic layer explicitly modeled as a "no-avoidance straight goer"—a setting made to isolate constructs, external validity limited.

Conversely, why does ordered sort motion fail at 0%? Because `nearest_target` always picks the nearest Euclidean target from the current position, while the task's waypoint order is

$A(-0.4, 1.3) \rightarrow B(-0.4, -1.3) \rightarrow C(1.6, 0.0)$, start $(-1.6, -1.3)$ nearest B. Motion first rushes B, but `_update_waypoints` strictly only recognizes active=0 A, hence motion forever stuck near B unable to enter the next target, finally failing as `max_steps` exhausted, score only 22.07 (only partial progress). Hybrid completes 100% because the semantic layer uses `_active_target` correctly choosing A, and takes over on 17% of decision segments to correct the prior's greedy drift.

These two contrasts together form this paper's core causal narrative: **complementarity is not "two weak models average stronger," but "two models each have structural blind spots on their construct dimension, hybrid lets the blind spot be covered by the other."** Intervention 4.7%'s meaning is therefore not "the semantic layer slacks," but "the semantic layer only acts on the 4.7% of decision segments where it is truly irreplaceable, trusting the prior the other 95.3%." This contrasts with the industry debate "frame-by-frame large-model backstop too costly": frame-by-frame backstop (direct) in this prototype both token-expensive and fails on contact tasks, while sparse takeover (hybrid) more economical and stable.

Appendix G Extended Analysis Directions (pre-submission route)

First, K scan. Currently fixed $K=32$. Theoretically larger K, candidate set more likely contains one "detour and order-preserving" good action, semantic takeover frequency should fall, but per-segment tokens ($\text{in}=900+40 \times K$) linearly rise, and motion prior compute cost rises. Expected on $K \in \{8, 16, 32, 64\}$ a "intervention—token—success" Pareto frontier, exactly H3's operationalization. Scan not run, curves recorded [RESULT NEEDED: K scan Pareto frontier].

Second, semantic error injection. Currently semantic critic a perfect rule (never errs within its construct). Real VLMs misjudge. Later injecting "wrong active target" or "misjudged collision" at controllable probability in the semantic layer, measuring intervention and success robustness to semantic noise, answering "when the semantic layer no longer perfect, is hybrid still better than pure prior."

Third, decision latency. Currently each segment decision instant on CPU, real VLMs have hundreds of ms to second latency. Later including "decision segment length $\times$ latency" in metrics, measuring long decision segments (using prior)'s actual benefit to control bandwidth. This especially key for the industry argument "sparse takeover reduces real-time pressure."

Fourth, token proxy calibration. Currently `TOKEN_PROXY` manual constants. After integrating a real VLM, replacing with returned usage fields, reporting "tokens per task / tokens per successful episode," upgrading directional conclusions to comparable cost numbers.

All four are post-gate work, this paper not pre-filling results.

Appendix H Further Note on Measurement Methodology and Metric Definitions

This paper holds "metrics must be frozen before data collection and mechanically produced by code," differing from the common "after-the-fact metric picking" in shared autonomy literature. Three core metrics' production as follows.

Success is an episode-level binary: the environment at `done` judges by `env.success` (all ordered targets reached and zero collisions), not back-inferred from a scoring threshold. This avoids the freedom of "tuning the score threshold to make success look good." Task score given by a fixed formula, including

reach fraction, progress fraction and no-collision bonus, its weights design parameters not after-the-fact fits. Intervention is a segment-level ratio: numerator `interventions` only increments when `chosen.kind=="override"`, denominator `decisions` records total decision segments, both mechanically accumulated in the main loop, no smoothing or subjective correction.

We especially stress intervention share vs "correction length." In external reports "modified 14.4% of control steps" may mean either "14.4% of steps overridden by semantic layer" or "the semantic layer's correction steps as a fraction of total steps," the two calibers not directly comparable. This paper uniformly uses the **decision-segment caliber**: a segment either wholly adopts the prior (6 steps) or wholly adopts the semantic correction (3 steps), intervention share the wholly adopted segments as a fraction of total. This caliber uniquely determined in code, readers recomputable from `raw_rows'` `decisions` and `interventions` columns, not needing to trust aggregates.

Token proxy production likewise transparent: each decision segment accumulated by mode looking up the `TOKEN_PROXY` dict, direct per segment 2200/650, hybrid 900+40K/260, motion zero. Since direct one segment per 3 steps, hybrid per 6, hybrid segment count naturally fewer; plus hybrid each segment only scores K compact candidates not frame-by-frame long reasoning, token directional decline structural, not a tuning result. Readers should read token numbers as "relative magnitude," not "absolute cost."

Appendix I Dialogue Boundary with Industry Debates

Recently industry has two extreme positions on "should large models directly control robots": one end holds that given a large enough model, frame-by-frame direct control generalizes to all contact tasks; the other holds large models expensive and unreliable, should retreat to pure planning, low level entirely to dedicated policies. This paper's CPU prototype provides a measurable middle-position evidence between the two: **pure semantic structurally fails on contact tasks, pure prior structurally fails on order tasks, while sparse hybrid lets each do its part.**

But we are clear about this evidence's boundaries. It cannot prove "hybrid beats end-to-end on all tasks," because: first, our semantic layer is a rule not a general large model, real large models' contact generalization may be far stronger than the rule critic; second, our prior is heuristic sampling not a learned diffusion policy, real priors may be more robust; third, the environment is a known-model point mass, real contact dynamics complexity tests both. Hence this paper's conclusion should be stated as "in a controlled synthetic environment, the value of complementarity and sparse takeover is reproducible and measurable," not "hybrid proven superior to end-to-end." This restraint is the fundamental difference from reports declaring architecture victory with a single success number.

Appendix J Reproducibility Statement and Author Division

All quantitative results here from fixed-seed deterministic computation. `torch.Generator` in `MotionPrior` initialized by `seed`, candidate sampling sequences reproducible; environment integration randomness-free deterministic; the only "external" data is the third-party report in the `external_reference` field, isolated outside results, entering no aggregate computation. Readers running Appendix A commands on the same tag should get byte-identical JSON (except timestamps). Author division and AI assistance see the end "Author Intended Statements."

Appendix K Summary

Synthesizing this paper's measurements: in the controlled CPU prototype, the architecture "semantic layer judges what to do, motion prior does it stably, semantic layer sparsely takes over at key points"

simultaneously improves success, collision safety and LLM call magnitude. Intervention share as a first-class metric makes this architecture no longer just engineering intuition, but a falsifiable proposition precisely measurable, adjustable by task difficulty, traded off by token cost. We look forward, after unlocking real VLMs and contact simulation, to using the same metric framework to test whether these conclusions hold on real robots; if complementarity then disappears or the intervention distribution changes, this paper's hypotheses will be correspondingly falsified—exactly the value of pre-registered experimental design.